H

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

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|---|---|---|---|
| CANDIDATE NAME | | | |
| CLASS | | TUTOR'S NAME | |
| CENTRE NUMBER | S | INDEX NUMBER | |

# PHYSICS 9478/03

Paper 3 Longer Structured Questions **21 September 2026**

**2 hours**

Candidates answer on the Question Paper.

No Additional Materials are required.

**READ THESE INSTRUCTIONS FIRST**

Write your name, class, Centre number and index number in the spaces at the top of this page.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams, graphs.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

**Section A**
Answer **all** questions.

**Section B**
Answer **one** question only.

You are advised to spend one and a half hours on Section A and half an hour on Section B.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| **Section A** | |
| **1** | **/ 7** |
| **2** | **/ 8** |
| **3** | **/ 7** |
| **4** | **/ 8** |
| **5** | **/ 9** |
| **6** | **/ 6** |
| **7** | **/ 10** |
| **Section B** | |
| **8** | **/ 20** |
| **9** | **/ 20** |
| **Total** | **/ 75** |

This document consists of **26** printed pages.

**Data**

| | | | |
|---|---|---|---|
| speed of light in free space | $c$ | = | $3.00 \times 10^{8}$ m s$^{-1}$ |
| permeability of free space | $\mu_0$ | = | $4\pi \times 10^{-7}$ H m$^{-1}$ |
| permittivity of free space | $\varepsilon_0$ | = | $8.85 \times 10^{-12}$ F m$^{-1}$ |
| | $(\frac{1}{4\pi\varepsilon_0}$ | = | $8.99 \times 10^{9}$ m F$^{-1}$) |
| elementary charge | $e$ | = | $1.60 \times 10^{-19}$ C |
| the Planck constant | $h$ | = | $6.63 \times 10^{-34}$ J s |
| unified atomic mass constant | $u$ | = | $1.66 \times 10^{-27}$ kg |
| rest mass of electron | $m_e$ | = | $9.11 \times 10^{-31}$ kg |
| rest mass of proton | $m_p$ | = | $1.67 \times 10^{-27}$ kg |
| molar gas constant | $R$ | = | 8.31 J K$^{-1}$ mol$^{-1}$ |
| Avogadro constant | $N_A$ | = | $6.02 \times 10^{23}$ mol$^{-1}$ |
| Boltzmann constant | $k$ | = | $1.38 \times 10^{-23}$ J K$^{-1}$ |
| gravitational constant | $G$ | = | $6.67 \times 10^{-11}$ N m$^2$ kg$^{-2}$ |
| acceleration of free fall | $g$ | = | 9.81 m s$^{-2}$ |

**Formulae**

| | | | |
|---|---|---|---|
| uniformly accelerated motion | $s$ | = | $ut + \frac{1}{2}at^2$ |
| | $v^2$ | = | $u^2 + 2as$ |
| work done on / by gas | $W$ | = | $p\,\Delta V$ |
| pressure | $p$ | = | $\frac{F}{A}$ |
| gravitational potential | $\phi$ | = | $-\frac{GM}{r}$ |
| temperature | $T / K$ | = | $T / °C + 273.15$ |
| pressure of an ideal gas | $p$ | = | $\frac{1}{3}\frac{Nm}{V} < c^2 >$ |
| mean translational kinetic energy of an ideal gas particle | $E$ | = | $\frac{3}{2}kT$ |
| displacement of particle in s.h.m. | $x$ | = | $x_0 \sin \omega t$ |
| velocity of particle in s.h.m. | $v$ | = | $v_0 \cos \omega t = \pm\omega\sqrt{\left(x_0^{\,2} - x^2\right)}$ |
| electric current | $I$ | = | $Anvq$ |
| resistors in series | $R$ | = | $R_1 + R_2 + \ldots$ |
| resistors in parallel | $1/R$ | = | $1/R_1 + 1/R_2 + \ldots$ |
| capacitors in series | $1/C$ | = | $1/C_1 + 1/C_2 + \ldots$ |
| capacitors in parallel | $C$ | = | $C_1 + C_2 + \ldots$ |
| energy in a capacitor | $U$ | = | $\frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$ |
| charging a capacitor | $Q$ | = | $Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$ |
| discharging a capacitor | Q | = | $Q_0 e^{-\frac{t}{\tau}}$ |
| RC time constant | $\tau$ | = | $RC$ |
| electric potential | $V$ | = | $\frac{Q}{4\pi\varepsilon_0 r}$ |
| alternating current / voltage | $x$ | = | $x_0 \sin \omega t$ |
| magnetic flux density due to a long straight wire | $B$ | = | $\frac{\mu_0 I}{2\pi d}$ |
| magnetic flux density due to a flat circular coil | $B$ | = | $\frac{\mu_0 NI}{2r}$ |
| magnetic flux density due to a long solenoid | $B$ | = | $\mu_0 nI$ |
| energy states for quantum particle in a box | $E_n$ | = | $\frac{h^2}{8mL^2}n^2$ |
| radioactive decay | $x$ | = | $x_0 e^{-\lambda t}$ |
| radioactive decay constant | $\lambda$ | = | $\frac{\ln 2}{t_{\frac{1}{2}}}$ |

## Section A

Answer **all** the questions in the spaces provided.

**1** Two spheres A and B of masses 3.7 kg and 2.7 kg respectively are attached to two light inextensible strings $L_1$ and $L_2$ each of length 0.50 m.

The combination of masses is made to swing in a vertical circle with both strings always in a straight line, as shown in Fig. 1.1.

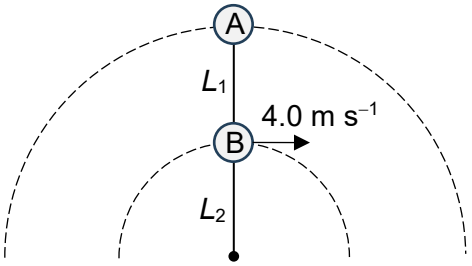



**Fig 1.1**

At the top of its motion, mass B is travelling at 4.0 m s$^{-1}$.

**(a)** Determine the speed of mass A at this instant.

speed of mass A = ..... m s$^{-1}$ [2]

**(b)** Show that the tension in $L_1$ at this instant is 200 N**.**

[1]

**(c)** Hence or otherwise, determine the tension in $L_2$ at this instant.

tension in $L_2$ = ..... N [2]

**(d)** At another instant in time, the combination of masses is at the bottom of its motion as shown in Fig. 1.2.

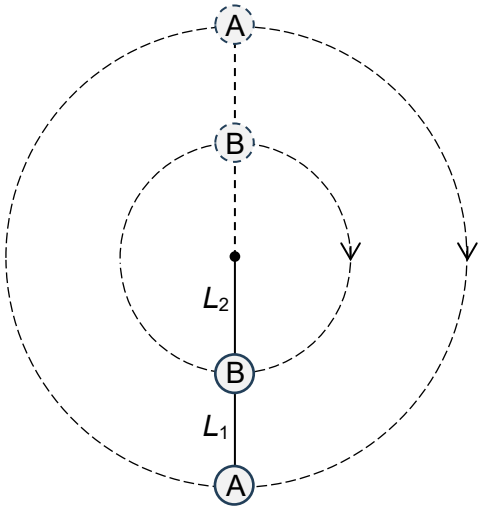



**Fig 1.2**

A student suggests that because the tensions in strings $L_1$ and $L_2$ change continuously during the motion, the total mechanical energy (i.e. kinetic and potential energy) of the system is not conserved.

With reference to tensions in the strings, state whether you agree with this statement and explain your reasoning.

.....

.....

.....

.....[2]

[Total: 7]

**2** **(a)** Define gravitational potential at a point

.....

.....

.....[2]

**(b)** **(i)** The dwarf planet Ceres may be considered to be an isolated uniform sphere of mass $9.4 \times 10^{20}$ kg and radius $4.7 \times 10^{5}$ m.

A particle is launched vertically upwards from the surface of the planet. Assuming negligible resistance from the planet's atmosphere, determine the minimum speed of this particle that will result in it escaping from the gravitational pull of the planet Ceres.

speed = ..... m $s^{-1}$ [3]

**(ii)** Hydrogen may be assumed to be an ideal gas.
The mass of a hydrogen molecule is $3.3 \times 10^{-27}$ kg.

Calculate the root-mean-square (r.m.s.) speed of a hydrogen molecule in hydrogen gas that is at a temperature of 240 K.

r.m.s. speed = ..... m $s^{-1}$ [2]

**(c)** The surface of planet Ceres reaches temperatures of approximately 240 K when in direct sunlight.

Use your answers in **(b)(i)** and **(b)(ii)** to suggest a reason why Ceres does not have an atmosphere consisting of hydrogen.

.....

.....

.....[1]

[Total: 8]

**3** A fixed amount of an ideal gas of mass 0.35 kg has pressure $2.0 \times 10^5$ Pa and volume 0.038 $m^3$.

**(a)** In one process, the gas is heated at constant pressure so that its internal energy increases by 7600 J, volume increases by 0.025 $m^3$ and temperature increases by 56°C.

**(i)** Determine the thermal energy $Q$ transferred to the gas.

$Q$ = ..... J [2]

**(ii)** Calculate the specific heat capacity of the gas for this process. Give a unit with your answer.

specific heat capacity = ..... unit ..... [2]

**(b)** In another process, the gas is heated at constant volume so that the increase in internal energy of the gas is the same as in **(a)**.

Use the first law of thermodynamics to explain whether the specific heat capacity of the gas for this process is less than, the same as, or greater than the answer in **(a)(ii)**.

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.....[3]

[Total: 7]

**4** **(a)** Fig. 4.1 shows a circular coil of 500 turns and radius 0.12 m.

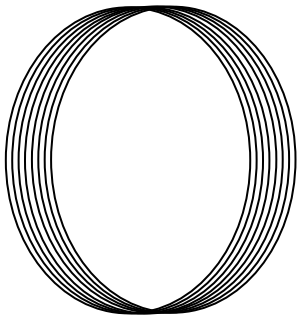


**Fig. 4.1**

A varying magnetic field is applied at right angles to the plane of the coil. Fig. 4.2 shows the variation with time *t* of this magnetic field of flux density *B*.

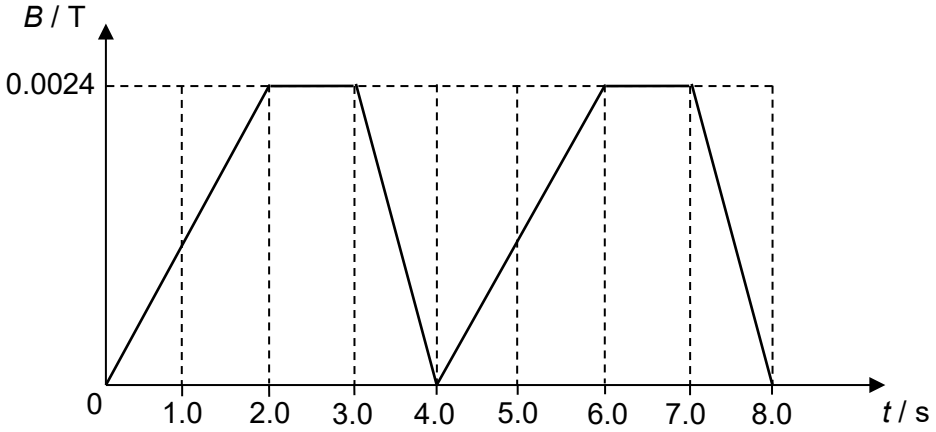



**Fig. 4.2**

**(i)** On Fig. 4.3, sketch a graph to show the variation with time t of the induced e.m.f. *E* in the coil. Include values on the y axis.

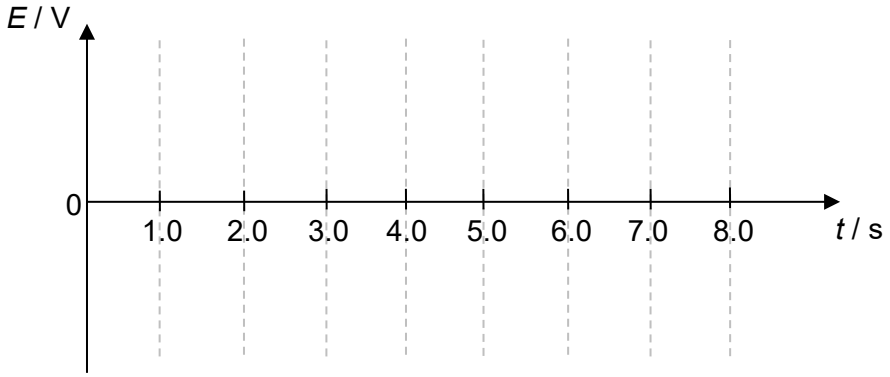



**Fig. 4.3**

[3]

**(ii)** Hence, determine the r.m.s value of the induced e.m.f.

r.m.s. value of induced e.m.f. = ..... V [2]

**(iii)** The e.m.f. source that you have sketched in Fig. 4.3 is now connected in series with a resistor *R* and an ideal diode as shown in Fig. 4.4.

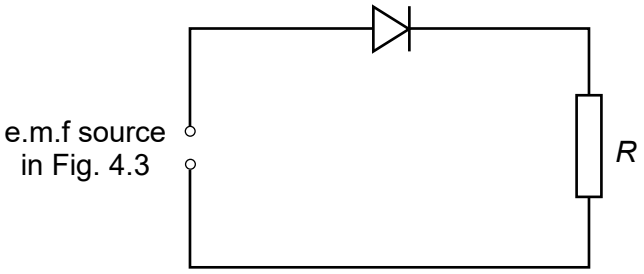



**Fig. 4.4**

State and explain the potential difference across R when the ideal diode has maximum potential difference across it.

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.....

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.....[3]

[Total: 8]

**5** **(a)** A straight rigid conducting wire of mass 0.14 kg and length 0.40 m is placed horizontally on a frictionless plastic plane. The plane is at an angle of 20° to the horizontal and is placed in a region of upwards uniform vertical magnetic field as shown in Fig. 5.1 and Fig. 5.2.

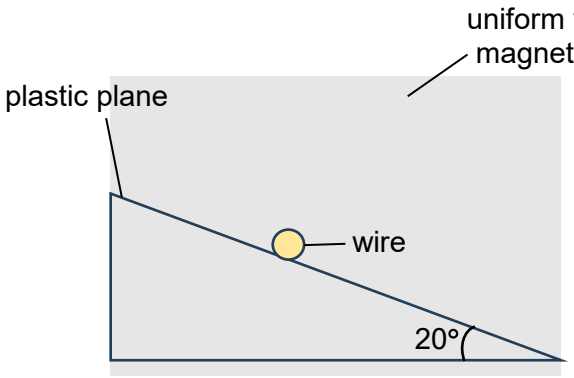



**Fig. 5.1 (side view)**

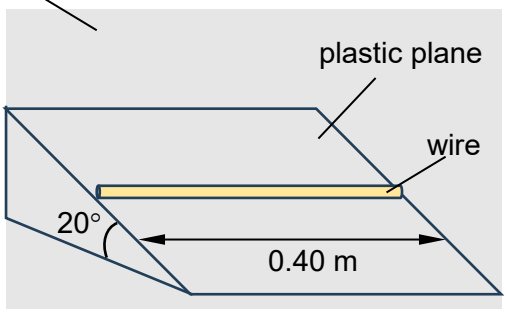



**Fig. 5.2**

To prevent the wire from sliding down the plane, a steady current $I$ is passed through the wire so that it remains stationary on the plastic plane.

**(i)** On Fig. 5.1, draw an arrow to show the direction of the magnetic force acting on the wire. [1]

**(ii)** On Fig. 5.2, draw an arrow to show the direction of the current $I$ in the wire. [1]

**(iii)** With reference to forces acting on the wire, explain why the wire remains stationary.

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.....[2]

**(iv)** Suggest and explain whether the wire will eventually slide up, remain stationary or slide down the plane if an alternating current of root-mean-square value of $I$ is passed through the wire instead.

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.....

.....[2]

**(b)** The current in **(a)** is disconnected, and the magnetic field is removed. The wire is now given a uniform net positive charge of $+2.0 \times 10^{-6}$ C.

**(i)** To keep the wire stationary in the same position on the frictionless plane, a uniform electric field of strength $E$ is applied in the horizontal direction.

Determine the magnitude of $E$.

$E$ = ..... N $C^{-1}$ [3]

[Total: 9]

**6** The pick-up on an electric guitar produces an electrical signal from the vibrations of the guitar string. The pick-up consists of a small coil of insulated wire wound round a small cylindrical magnet as shown in Fig. 6.1.

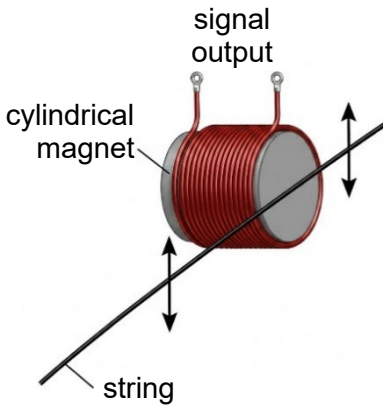



**Fig. 6.1**

The strings of the guitar are made of steel, which is a conductor. When a string vibrates, an electrical signal is generated between the terminals of the coil.

**(a)** Use Faraday's Law to explain why an electrical signal is generated.

.....

.....

.....

.....

.....[3]

**(b)** Suggest why

**(i)** the electrical signal needs to be amplified before connection to a loudspeaker,

.....

.....[1]

**(ii)** the design of the pick-up would be inappropriate for use on a guitar with nylon strings.

.....

.....

.....

.....[2]

[Total: 6]

**7** **(a)** An electron is confined inside a one-dimensional potential well of length 0.50 nm with infinitely high potential barriers at $x = 0$ and $x = 0.50$ nm.

Within the well, the potential energy $U(x) = 0$, and outside the well, $U(x) = \infty$.

**(i)** By considering the de Broglie wavelength of the electron trapped in this well, explain why the electron has specific allowed energy levels.

.....

.....

.....

.....

.....[2]

**(ii)** The electron emits a single photon when it transitions from the second excited state to the ground state.

Determine the wavelength of the photon.

wavelength = ..... m [3]

**(iii)** Explain, with reference to the Heisenberg Uncertainty Principle, why the ground state energy of the electron cannot be zero.

.....

.....

.....

.....[2]

**(b)** An electron bound within a hydrogen atom has allowed energy levels shown in Fig. 7.1.

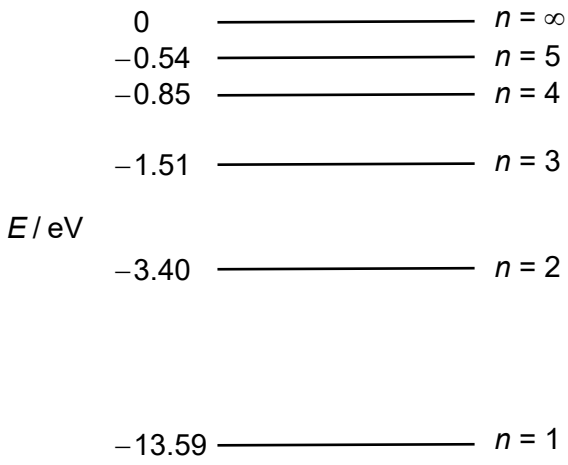



**Fig. 7.1**

**(i)** Suggest one reason why the energy levels of this electron are different from those of the electron in **(a)**.

.....

.....

.....[1]

**(ii)** The electron emits a single photon when it transitions from the second excited state to the ground state. Determine the wavelength of the photon.

wavelength = ..... m [2]

[Total: 10]

## Section B

Answer **one** question from this section in the spaces provided.

**8** **(a)** A small steel sphere is oscillating vertically on a spring, as shown in Fig. 8.1.

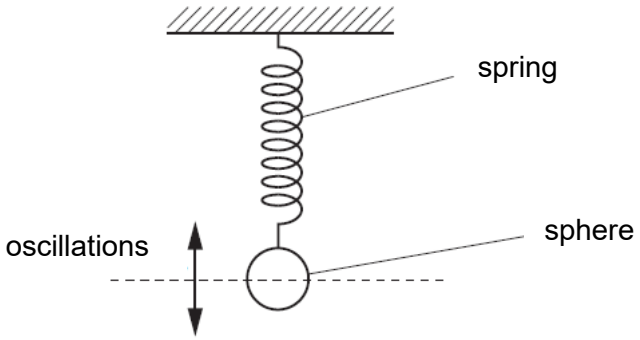



**Fig. 8.1**

Fig. 8.2 shows the variation with displacement $x$ of the energy $E$ of the oscillations.

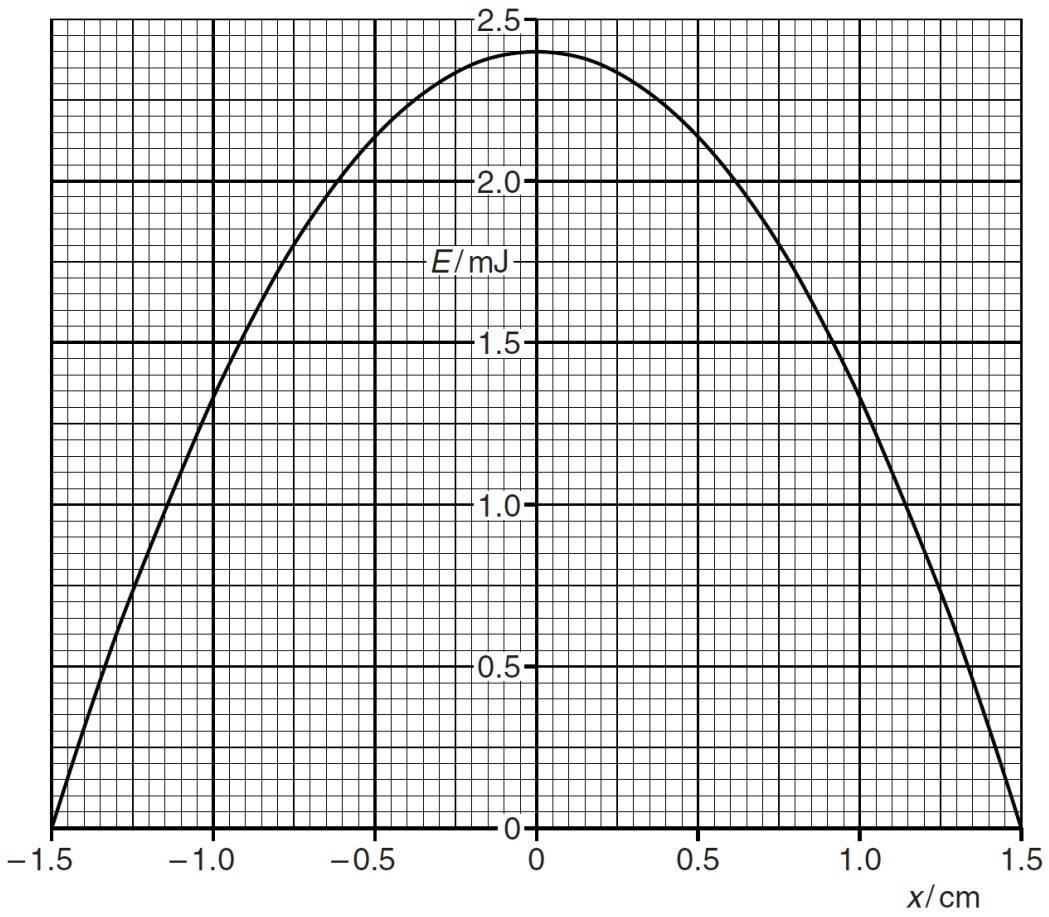



**Fig. 8.2**

**(i)** State whether the energy $E$ represents the total energy, the potential energy or the kinetic energy of the oscillations.

.....[1]

**(ii)** The amplitude of the oscillations reduces over a long period of time. After many oscillations, the maximum value of $E$ decreases **by** 0.60 mJ.

Without further calculation, use Fig. 8.2 to determine the new amplitude of the oscillations. Show your construction.

new amplitude = ..... cm [2]

**(b)** The steel sphere is held between two fixed points A and B by two identical springs, as shown in Fig. 8.3.

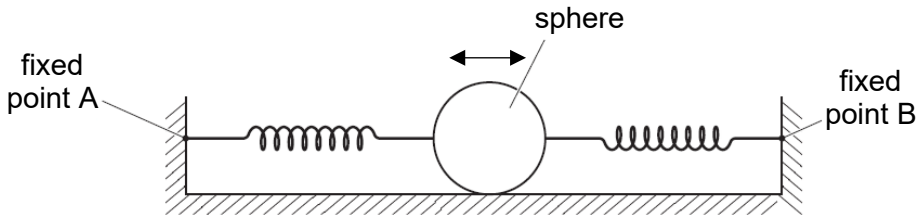



**Fig. 8.3**

The sphere is displaced from its equilibrium position and then released so that it is free to oscillate horizontally along the line AB. During the oscillations, the springs remain extended and do not exceed their limits of proportionality.

The resultant force acting on the sphere is given by

$$F = -2kx$$

where $k$ is the spring constant and $x$ is the displacement of the sphere from equilibrium position.

**(i)** Explain how the equation shows that the oscillations of the sphere are simple harmonic.

.....

.....

.....

.....

.....[2]

**(ii)** Show that the frequency $f$ of the oscillations is given by

$$f = \frac{1}{2\pi}\sqrt{\frac{2k}{m}}$$

where $m$ is the mass of the sphere.

[2]

The variation of the acceleration $a$ of the sphere with its displacement $x$ from the equilibrium position is shown in Fig. 8.4.

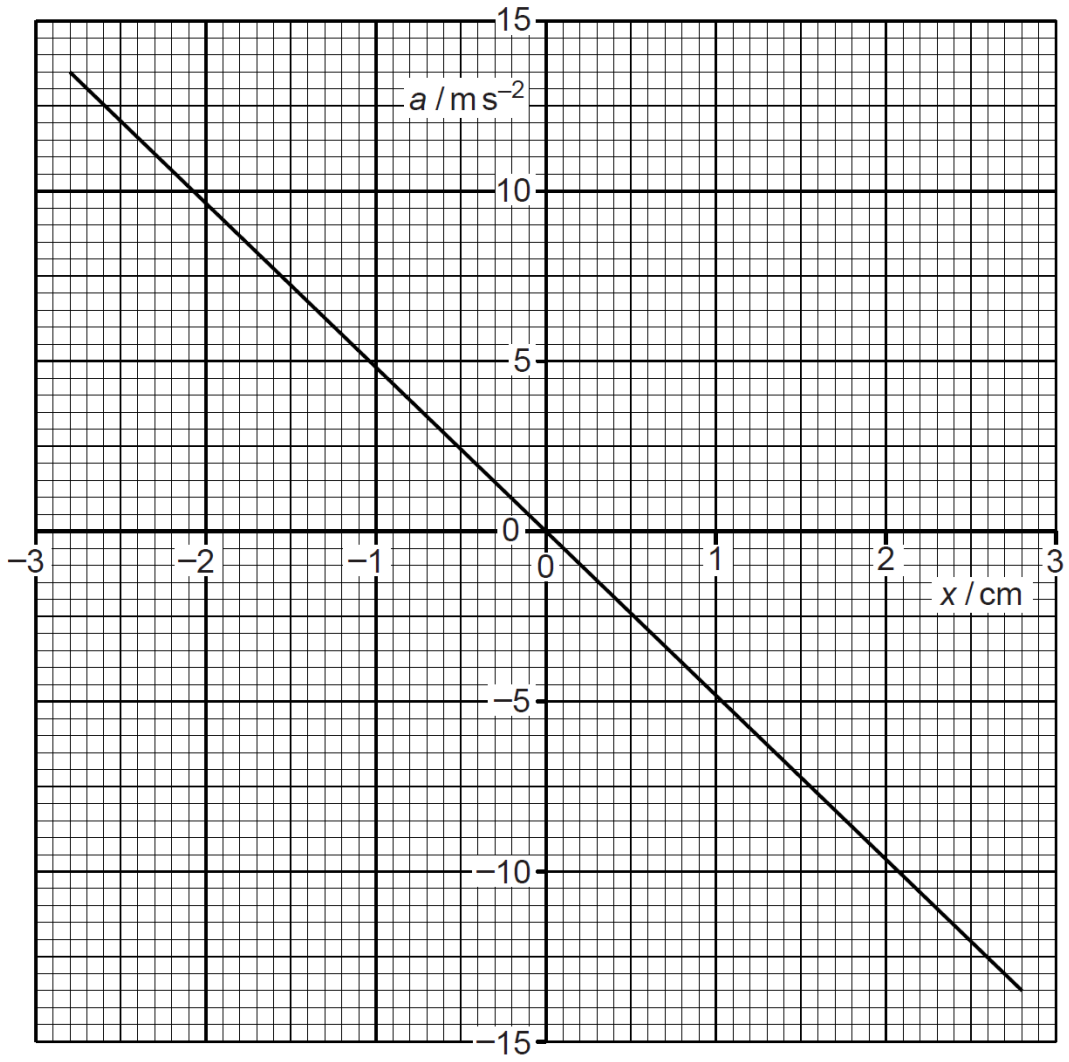



**Fig. 8.4**

**(iii)** Use Fig. 8.4. to determine the frequency of oscillations of the sphere.

$f$ = ..... Hz [3]

**(iv)** State **one** way how the period of oscillations of the sphere can be reduced.

.....

.....[1]

**(v)** The variation of the displacement of the sphere with time is sinusoidal. The variation of the velocity of the sphere with time is also sinusoidal.

**1.** State the phase difference between the displacement and the velocity of the sphere.

phase difference = .....° [1]

**2.** The sphere is released from its maximum positive displacement at time $t = 0$.

Calculate the magnitude of the velocity of the sphere at time $t = \frac{T}{12}$, where $T$ is the period of oscillations.

velocity = ..... m $s^{-1}$ [3]

**(vi)** The microscopic structure of a solid crystal can be pictured as planes of atoms, as shown in Fig. 8.5. The distance between the centres of neighbouring atoms is indicated.

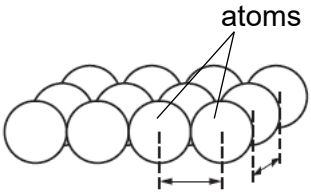



**Fig. 8.5**

Suggest why the spring-sphere system in Fig. 8.3 provides a simple model for the motion of an atom in a solid crystal.

.....

.....

.....[1]

**(c)** The fixed point B is now replaced with an oscillator that causes horizontal oscillations of the sphere, as shown in Fig. 8.6. Both springs remain extended within the limit of proportionality.

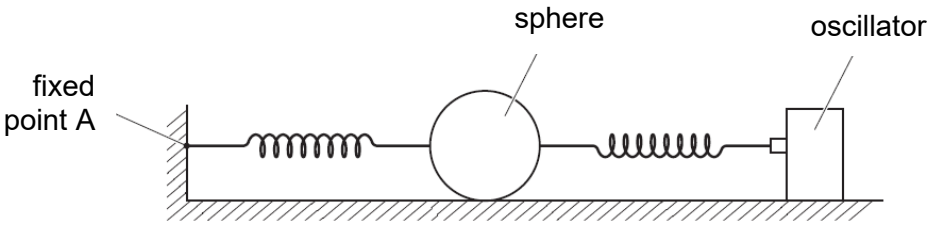



**Fig. 8.6**

The oscillator is switched on. The amplitude of vibration of the oscillator is constant. The frequency $f$ of vibration of the oscillator is varied.

The variation of the amplitude $x_0$ of the oscillations with frequency $f$ of the oscillator is shown in Fig. 8.7.

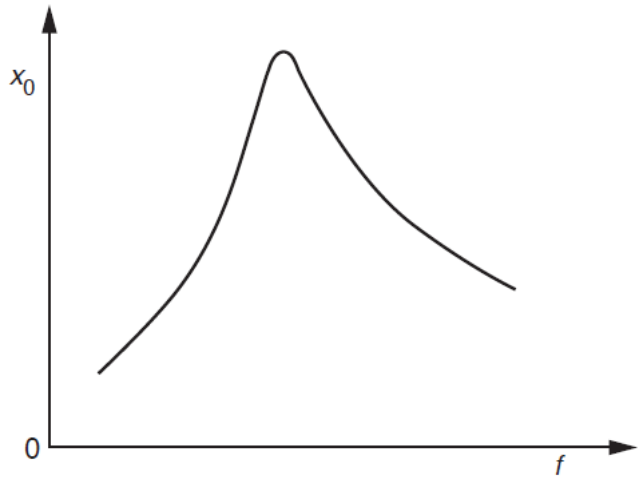



**Fig. 8.7**

**(i)** Explain why there is a peak in Fig. 8.7 and state the frequency at which the peak occurs.

.....

.....

.....

.....[2]

**(ii)** State and explain **one** modification to the setup in Fig. 8.6 to reduce the amplitude of the oscillations in Fig. 8.7.

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.....

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.....[2]

[Total: 20]

**9** **(a)** In a circuit, a 4.0 V battery with internal resistance is connected in series to a thermistor and a coaxial resistor.

The coaxial resistor has a cylindrical core of length 0.020 m. The core is made of material A and is surrounded by material B as shown in Fig. 9.1. In the circuit, the direction of the current passing through the coaxial resistor is parallel to its length. The physical dimensions of the coaxial resistor are fixed.

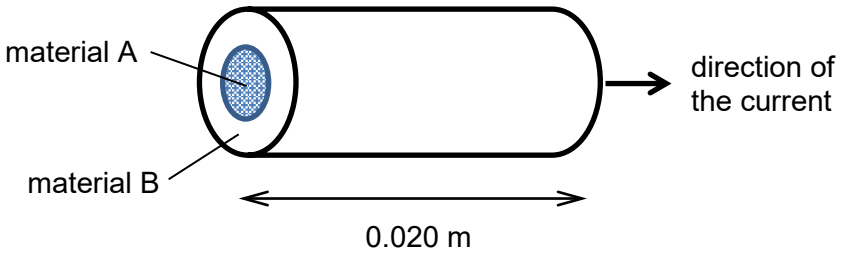



**Fig. 9.1**

**(i)** Draw a complete circuit diagram of the circuit in the space below.

Represent the coaxial resistor as two separate resistors A and B, one for each material respectively.

[3]

**(ii)** Fig. 9.2 shows the variation of the current $I$ in the thermistor with temperature $T$ of the thermistor.

Fig 9.3 shows the variation of $I$ in the thermistor with voltage $V$ across the thermistor.

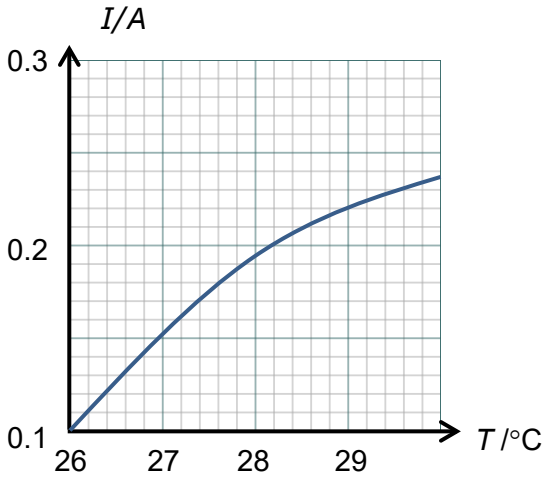



**Fig. 9.2**

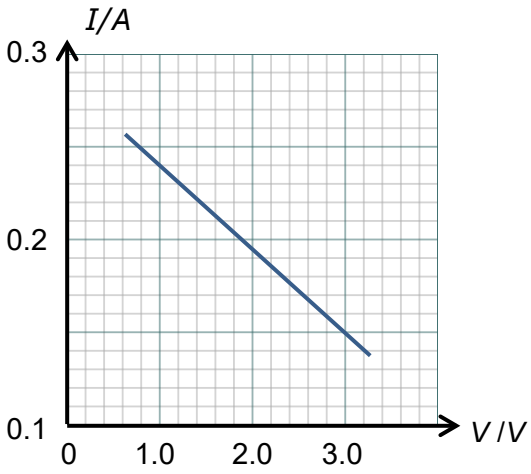



**Fig. 9.3**

Determine the resistance of the thermistor at 27 °C.

resistance = ..... Ω [2]

**(iii)** Describe and explain what will happen to the potential difference across resistors A and B in the circuit when the temperature of the thermistor increases.

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.....[2]

**(b) (i)** Two identical lamps are connected to a d.c. and a.c. supply as shown in Fig. 9.4a and Fig. 9.4b respectively.

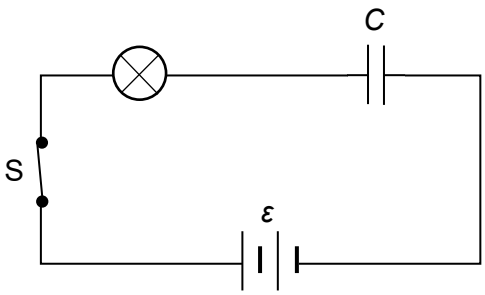



**Fig 9.4a**

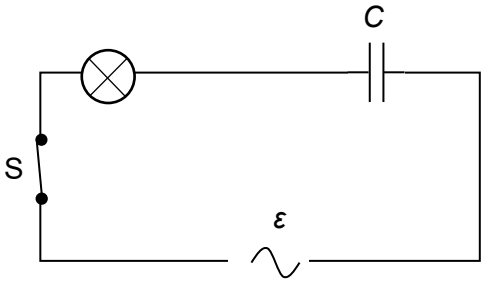



**Fig 9.4b**

State and explain which lamp will be turned on for a longer period of time. Assume that the time constant is large.

.....

.....

.....

.....

.....[3]

**(ii)** The charged capacitor of capacitance *C* is being discharged through a resistor of resistance *R*, as shown in Fig. 9.5. At time = 0 s, the capacitor is fully charged.

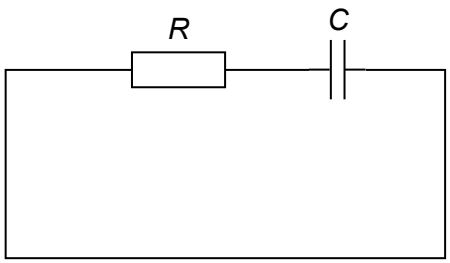



**Fig. 9.5**

The energy stored in the capacitor decreases with time as the capacitor discharges. Determine the time $t$, in terms of $R$ and $C$, it takes for this stored energy to be discharged by 75% of its initial value.

$t$ = ..... [3]

**(c)** After a long time, the capacitor in Fig. 9.5 is fully discharged and a battery of e.m.f $E$ of negligible internal resistance is used to recharge the capacitor in a circuit as shown in Fig. 9.6.

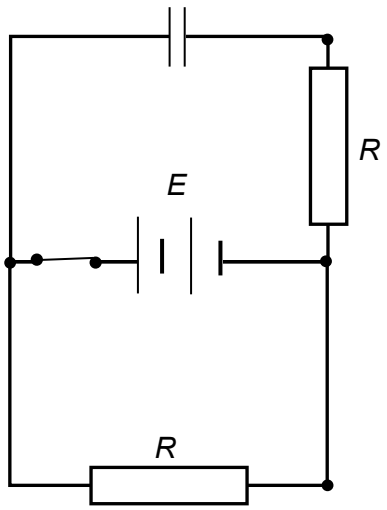



**Fig. 9.6**

**(i)** Determine the current, in terms of $E$ and $R$, in the battery immediately after the switch is closed.

current = ..... [2]

**(ii)** Determine the current, in terms of $E$ and $R$, in the battery a long time after the switch is closed.

current = ..... [2]

**(iii)** Hence suggest an expression for the time constant, in terms of $R$ and $C$.

Explain your reasoning.

.....

.....

.....

.....

.....

.....[3]

[Total: 20]

**END OF PAPER**